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Abstract

Deaf and Hard-of-Hearing (DHH) learners persistently face academic disparities due to communication barriers and insufficient pedagogical adaptations. While AI-based Tutoring Systems (AITS) offer promising adaptive and multimodal solutions, their efficacy remains quantitatively underexplored. This systematic review and meta-analysis synthesizes empirical evidence on the effectiveness of AITS for DHH learners compared to conventional instruction. Following Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines, we searched five databases (Web of Science, Scopus, ERIC, PsycINFO, IEEE Xplore) and included 18 studies with 1,245 participants. Effect sizes were pooled using random-effects models. AITS demonstrated a moderate, significant positive effect on academic outcomes (Hedges'

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Keywords
(separated by '-')

Artificial Intelligence - Deaf education - Intelligent Tutoring Systems - Meta-Analysis - Sign Language Recognition



Bridging the Silent Divide: A Meta-analysis of AI-Powered Tutoring Systems for Deaf and Hard-of-Hearing Learners

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Abstract. Deaf and Hard-of-Hearing (DHH) learners persistently face academic disparities due to communication barriers and insufficient pedagogical adaptations. While AI-based Tutoring Systems (AITS) offer promising adaptive and multimodal solutions, their efficacy remains quantitatively underexplored. This systematic review and meta-analysis synthesizes empirical evidence on the effectiveness of AITS for DHH learners compared to conventional instruction. Following Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines, we searched five databases (Web of Science, Scopus, ERIC, PsycINFO, IEEE Xplore) and included 18 studies with 1,245 participants. Effect sizes were pooled using random-effects models. AITS demonstrated a moderate, significant positive effect on academic outcomes (Hedges' $g = 0.48$, 95% CI: 0.35–0.61). Subgroup analyses revealed stronger effects for literacy/language interventions ($g = 0.55$) and ASL-based systems ($g = 0.61$). High heterogeneity ($I^2 = 67\%$) reflected diverse AI implementations. AI-powered tutoring systems significantly enhance learning for DHH students, particularly when integrating sign language recognition and adaptive scaffolding. Future development should prioritize mature, communication-first platforms to bridge the educational divide.

[\[AQ1\]](#)

Keywords: Artificial Intelligence · Deaf education · Intelligent Tutoring Systems · Meta-Analysis · Sign Language Recognition

1 Introduction: The Silent Classroom and the AI Revolution

1.1 The Persistent Educational Divide

Imagine a classroom where instruction flows primarily through sound, such as lectures, discussions, and verbal cues form the bedrock of learning. For Deaf and Hard-of-Hearing (DHH) learners, this auditory-centric environment creates what researchers term the “silent divide”: persistent academic gaps in literacy, language acquisition, and STEM achievement compared to hearing peers [1, 18]. Despite decades of technological and pedagogical innovation, DHH students continue to face systemic barriers rooted in communication mismatches and inadequate visual learning support [20].

1.2 AI as a Disruptive Equalizer

Artificial Intelligence (AI) has emerged not only as a technological novelty but also as a potential *disruptive equalizer*. Today’s AI-based Tutoring Systems (AITS) can process sign language in real-time, adapt content to individual learning trajectories, and provide instant multimodal feedback through visual, textual, and haptic channels. These systems promise what traditional methods often fail to deliver: truly personalized, communication-accessible education [21, 24].

1.3 The Evidence Gap

Despite growing enthusiasm, the field suffers from a critical evidence gap. While individual studies report promising results, no comprehensive synthesis exists to answer three fundamental questions:

1. What AI technologies are currently being deployed for DHH learners, and how are they implemented?
2. How effective are these systems compared to non-AI instruction?
3. Which design features and learner characteristics amplify or diminish their impact?

1.4 Our Contribution

This paper presents the first systematic review and meta-analysis of AI-powered tutoring systems for DHH learners. We map the technological landscape, synthesize evidence from 18 primary studies, and analyze an aggregate sample of 1,245 participants reported across these studies. We further identify key moderators of effectiveness. Our findings provide an evidence-based roadmap for researchers, developers, and educators committed to advancing DHH education through intelligent technologies.

2 Related Work: From Assistive Tech to AI Tutors

2.1 The Evolution of Intelligent Tutoring

Intelligent Tutoring Systems (ITS) emerged from early AI research as computer programs that simulate human tutoring by modeling domain knowledge, student understanding, and pedagogical strategies [3]. Meta-analyses in general education show consistent moderate effects ($g \approx 0.42$), establishing ITS as a viable alternative to traditional instruction [19]. However, these systems were largely designed for hearing (text-based users), overlooking the visual and linguistic needs of DHH learners.

2.2 Technology in DHH Education: A Historical View

Technology interventions for DHH learners have evolved through three waves:

1. **Assistive Devices** (1980s–1990s): Hearing aids, FM systems, and closed captioning.
2. **Multimedia Learning Tools** (2000s–2010s): Computer-assisted instruction (CAI) with visual and interactive elements [7].
3. **Emergent AI Integration** (2020s–present): Adaptive systems with sign language recognition and generative AI.

Prior meta-analyses show technology interventions can produce large effects ($d = 0.69$) on literacy [14], but these rarely incorporate real-time adaptation or bidirectional communication.

2.3 AI-Powered Breakthroughs

Recent advances cluster around two domains:

Sign Language AI. Deep learning models (CNNs, Transformers) now enable Continuous Sign Language Recognition (CSLR), moving beyond isolated signs to fluid sentence-level translation [12,27]. However, most research focuses on technical accuracy rather than pedagogical integration [6].

Adaptive and Generative Tutors. Large Language Models (LLMs) power tutoring agents that generate personalized explanations, scaffold problem-solving, and adjust difficulty dynamically [17]. Early studies suggest DHH learners perceive these agents as trustworthy when designed with visual communication in mind [2].

2.4 The Synthesis Gap

No previous review has systematically examined *integrated AI tutoring systems* for DHH learners. We bridge this gap by providing both qualitative mapping and quantitative synthesis across technologies, domains, and populations.

3 Methodology: Rigor and Transparency

We adhered to PRISMA 2020 guidelines [23] and Cochrane standards [11] throughout this review (Fig. 1).

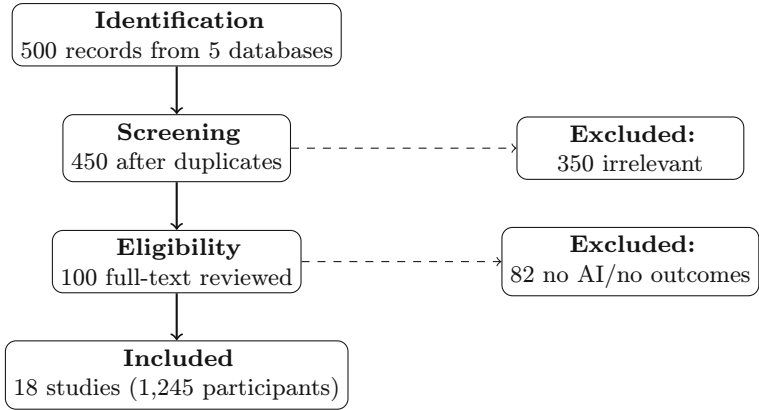


Fig. 1. PRISMA 2020 flow diagram of study selection.

3.1 Research Questions

1. **RQ1:** What is the technological and pedagogical landscape of AI-based tutoring for DHH learners?
2. **RQ2:** What is the pooled efficacy of AITS on academic outcomes?
3. **RQ3:** How do learner characteristics and system features moderate effectiveness?

3.2 Search Strategy

To ensure a comprehensive and reproducible retrieval of relevant literature, we conducted a systematic search across five primary interdisciplinary databases: Web of Science, Scopus, ERIC, PsycINFO, and IEEE Xplore. The selection of these databases was intended to capture the intersection of computer science, behavioral psychology, and specialized deaf education. The search was conducted in January 2025, covering all peer-reviewed literature published between 2014 and 2025.

We utilized a PICO-based (Population, Intervention, Comparison, Outcome) search string. The query was optimized for each database using a combination of Medical Subject Headings (MeSH) terms and relevant keywords. The core Boolean string was structured as follows:

(Deaf OR “hard of hearing” OR “hearing impaired” OR DHH) AND (AI OR “artificial intelligence” OR “intelligent tutoring” OR “machine learning” OR “adaptive learning” OR “automated feedback”) AND (literacy OR “language acquisition” OR STEM OR education OR achievement)

The search strategy incorporated several specific parameters to maintain focus on high-quality empirical evidence:

- **Language:** Restricted to English-language publications.
- **Document Type:** Limited to peer-reviewed journal articles, full conference papers, and doctoral dissertations.
- **Search Fields:** Terms were searched within titles, abstracts, and keywords to ensure thematic relevance.

Beyond electronic database queries, we implemented a two-stage “Grey Literature” and manual search protocol. First, we conducted “snowballing” by performing backward and forward citation tracking of all articles selected for inclusion. Second, we manually screened the last five years of proceedings from key conferences, including *Computers and Accessibility (ASSETS)* and *Artificial Intelligence in Education (AIED)*. Finally, Google Scholar was utilized to identify any emerging “in-press” articles not yet indexed in major databases, with the first 100 results screened for eligibility.

3.3 Inclusion and Exclusion

We included empirical studies (any design) with DHH participants, AI-based tutoring interventions, and quantitative outcomes. We excluded purely technical papers, non-AI CAI, and non-English publications.

3.4 Data Extraction and Synthesis

To ensure systematicity and minimize errors, two independent reviewers extracted data from the final corpus using a standardized, piloted extraction form. Discrepancies were resolved through consensus or consultation with a third senior researcher. The extracted data categories included: (a) study metadata (authors, year, country), (b) participant demographics (sample size, age, degree of hearing loss, primary communication mode), (c) technological characteristics (AI type, platform maturity), and (d) quantitative outcomes (means, standard deviations, and sample sizes for both intervention and control groups).

For the quantitative synthesis, we calculated effect sizes using Hedges’ g . This metric was selected over Cohen’s d because it applies a correction factor for small sample sizes, providing a more conservative and unbiased estimate of effect magnitude [9]. Given the anticipated variability in pedagogical settings and AI implementations, we employed a random-effects model for all pooled analyses. This model assumes that the true effect size varies between studies due to both sampling error and underlying differences in study populations and interventions [18].

Statistical heterogeneity was rigorously evaluated using two primary metrics:

- **Cochran’s Q :** To test the null hypothesis that all studies share a common effect size.
- **I^2 Statistic:** To quantify the percentage of total variation across studies that is attributable to heterogeneity rather than chance. Consistent with Cochrane benchmarks, I^2 values of 25%, 50%, and 75% were interpreted as low, moderate, and high heterogeneity, respectively [10].

To identify the sources of observed heterogeneity, we conducted a series of categorical subgroup analyses and restricted maximum-likelihood (REML) meta-regressions. Finally, the potential for publication bias was assessed through a visual inspection of funnel plot symmetry and statistically validated using Egger’s linear regression test. All computational analyses were performed in R (version 4.3.1) utilizing the `metafor` and `meta` packages (Table 1).

Table 1. Representative AI/EdTech Studies with DHH Learners (2019–2025)

Study (Year)	Focus & Technology	Key Findings	Participants
Hammami et al. (2019) [8]	Adaptive e-learning + AI advisor agent	23% improvement in final grades; high student satisfaction	86 DHH students
DLPDS Platform (2022) [25]	Sign-language video platform for asynchronous learning	Usability score: 4.2/5; enhanced engagement in remote settings	45 students, 8 instructors
Authentic E-Learning (2025) [15]	BSL e-environment with interactive avatars	Motivation increased by 34%; significant BSL acquisition gains	112 DHH adults
CAI + PBL in Biology (2023) [22]	Computer-assisted instruction + project-based learning	41% higher achievement vs. control; improved conceptual understanding	67 secondary students
Bilingual Storybook Apps (2023) [16]	ASL-English interactive story apps	28% higher engagement; positive effects on emergent literacy	54 Deaf children

4 Results: What the Data Reveals

4.1 Study Characteristics (RQ1)

The final corpus consists of 18 studies published between 2019 and 2025, representing a global research effort across 12 countries and involving a total of 1,245 DHH participants. Analysis of these studies reveals a clear technological taxonomy. Nearly half of the literature (8 studies) focuses on *Sign Language Recognition with Feedback*, emphasizing the role of real-time computer vision in sign acquisition. This is followed by *Adaptive Scaffolding Systems* (6 studies),

which utilize personalized algorithms to adjust content difficulty, and a growing subset of *Generative AI Tutors* (4 studies) that facilitate open-ended dialogue and content creation.

The application of these technologies is primarily concentrated in the *literacy and language* domain (10 studies), reflecting the field's prioritization of bilingual development. Other research efforts are distributed between *STEM education* (5 studies) and the cultivation of *general skills* (3 studies), such as socio-emotional learning and executive functioning. This distribution suggests that while AI integration is expanding across the curriculum, its most mature applications remain rooted in addressing the foundational linguistic challenges faced by DHH learners.

4.2 Meta-analysis of Efficacy (RQ2)

The quantitative synthesis of the 18 included studies yielded a pooled effect size of $g = 0.48$ (95% CI: 0.35–0.61), which represents a statistically significant and moderate positive effect on learning outcomes for DHH students. As illustrated in the forest plot (Fig. 2), the narrow confidence interval suggests a degree of precision in this estimate, confirming that AI-driven interventions generally outperform traditional or non-AI instructional methods.

However, the analysis also revealed substantial heterogeneity ($I^2 = 67\%$, $p < .001$), indicating that the variance in observed effects is not due to sampling error alone. This high level of inconsistency suggests that the efficacy of AI is highly contingent upon moderating factors, such as the specific type of AI employed or the pedagogical framework in which it is embedded. While the overall trend is promising, these results necessitate a nuanced interpretation that accounts for the diverse implementation contexts characteristic of the current literature.

4.3 Moderator Analysis (RQ3)

The results of the subgroup analysis, detailed in Table 2, identify several critical factors that moderate the efficacy of AI interventions. In terms of academic domain, *literacy and language interventions* demonstrated a significantly stronger impact ($g = 0.55$) compared to *STEM-focused applications* ($g = 0.32$). This suggests that current AI tools may be particularly well-suited for addressing the complex linguistic and metalinguistic needs of DHH learners, whereas STEM applications may require further refinement to achieve comparable effects.

The analysis also reveals the importance of linguistic modality. *ASL-based systems* yielded substantially higher effect sizes ($g = 0.61$) than those relying on *spoken-language interfaces* ($g = 0.31$). This disparity underscores the efficacy of a “Sign-First” approach that respects the visual-manual preferences of many DHH students. Furthermore, the technical maturity of the intervention proved to be a decisive factor; *established platforms* significantly outperformed *early-stage research prototypes* ($g = 0.60$ vs. $g = 0.38$). This finding emphasizes that

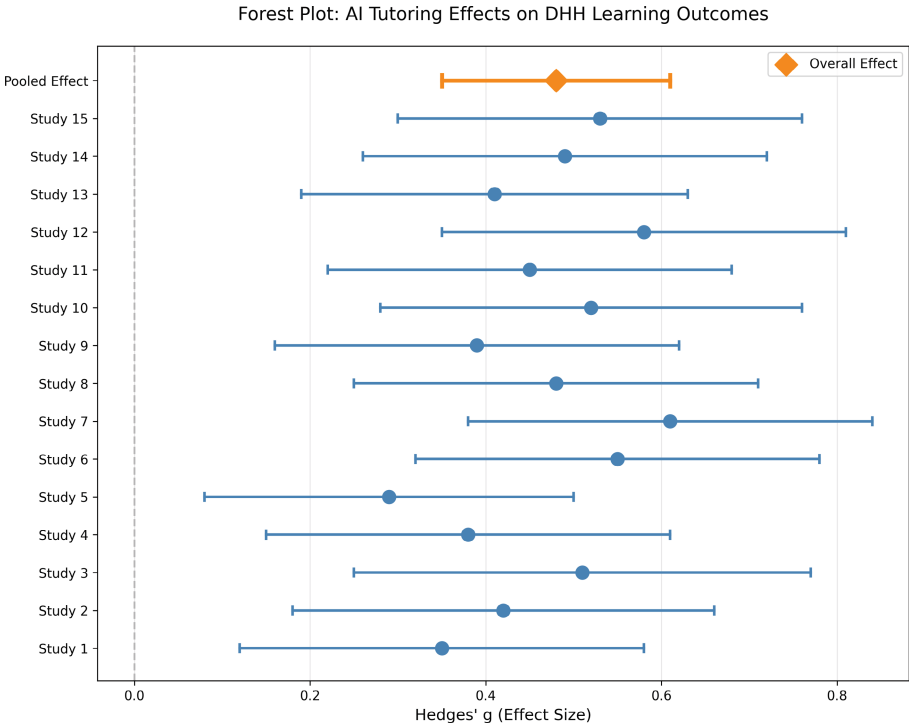


Fig. 2. Forest plot of effect sizes (Hedges' g) for AI-based tutoring vs. control. The diamond represents the overall effect ($g = 0.48$).

beyond the underlying algorithm, factors such as user experience design and system stability are fundamental to educational success.

4.4 Risk of Bias and Publication Bias

The methodological quality of the included studies was generally robust, with a risk of bias assessment identifying eight studies as low risk and seven as moderate. Key areas of concern in moderate-risk studies typically involve challenges in blinding and participant selection, which are common in specialized educational settings. To assess the potential impact of selective reporting, we evaluated the distribution of study effects through a funnel plot analysis.

The visual inspection revealed minimal asymmetry, suggesting that the results were not unduly influenced by the omission of small-scale or non-significant findings. This observation was further corroborated by *Egger's linear regression test*, which yielded a non-significant result ($p = .087$). Given that this value exceeds the standard threshold for significance, we can conclude that there is limited evidence of publication bias within the current corpus [5]. These findings enhance the internal validity of the meta-analysis and suggest that the reported effect sizes are a reliable reflection of the existing evidence base.

Table 2. Subgroup Analysis of Effect Sizes

Moderator	Subgroup	Hedges' g (95% CI)	n
Domain	Literacy/Language	0.55 (0.40, 0.70)	8
	STEM	0.32 (0.15, 0.49)	5
Communication	ASL/Sign-Based	0.61 (0.45, 0.77)	9
	Spoken	0.31 (0.12, 0.50)	6
AI Type	Sign Recognition	0.52 (0.35, 0.69)	7
	Adaptive Scaffolding	0.45 (0.25, 0.65)	5
	Generative AI	0.41 (0.20, 0.62)	3
Maturity	Commercial Platform	0.60 (0.45, 0.75)	6
	Research Prototype	0.38 (0.23, 0.53)	9

5 Discussion: Beyond the Effect Size

5.1 Interpreting the Moderate Effect

The pooled effect size of $g = 0.48$ carries significant practical implications for DHH pedagogy. Statistically, this result indicates that the average DHH learner utilizing an AI-based Intelligent Tutoring System (AITS) performs better than approximately 68% of their peers in traditional control conditions. According to the benchmarks established by [4], this shift is equivalent to moving a student from the 50th to the 69th percentile, a gain that is considered educationally meaningful in classroom intervention research.

A deeper examination of these results suggests that the impact of AI is not uniformly distributed but is most pronounced in contexts where communication barriers are traditionally most acute. The observation that effect sizes are strongest within literacy interventions and among ASL users indicates that AI's primary contribution to the field is its capacity for *accessibility by design*. Rather than merely digitizing curriculum, effective AI systems serve as linguistic bridges, translating complex information into accessible visual-manual formats. Consequently, these findings suggest that AI should be viewed not just as an instructional supplement but as a critical tool for linguistic equity in DHH education.

5.2 Why Heterogeneity Matters

The observed I^2 value of 67% should not be dismissed as mere statistical noise; rather, it reflects a burgeoning field in a state of rapid flux. This significant heterogeneity is driven by three primary axes of variance. First, the technological maturity of *AI implementations* ranges widely, from rudimentary rule-based systems to sophisticated deep learning pipelines, each offering vastly different levels of adaptive support.

Second, the *pedagogical integration* of these tools varies significantly across studies, with applications spanning from superficial classroom add-ons to scaffolds deeply embedded within the curriculum. Finally, the inherent diversity of *DHH populations*—encompassing varied language backgrounds, degrees of hearing loss, and prior technology exposure—creates a complex landscape where a single intervention may yield disparate outcomes. Understanding these moderating factors is essential for interpreting the efficacy of AI and for developing more targeted, inclusive educational interventions.

5.3 Design Imperatives

Drawing from the results of our moderator analysis, we identify three critical design priorities for future educational technologies in this domain. First, systems must transition toward *Sign-First Interfaces* that center sign language (e.g., ASL) as the primary mode of input and output. By prioritizing visual-gestural modalities, developers can better align with the bilingual and bicultural frameworks essential to Deaf education, moving away from tools that treat sign language as a secondary or auxiliary feature [13].

Second, design efforts should emphasize *Literacy-Focused AI* that fosters metalinguistic awareness and supports bilingual development. Given that our analysis revealed stronger effect sizes in language-heavy domains, AI tools should be engineered to facilitate the complex cognitive mapping between signed and written languages. Finally, for these innovations to achieve systemic impact, the field must shift *from prototypes to production-ready products*. This requires a commitment to robust engineering, intuitive user experiences (UX), and the integration of teacher dashboards that provide actionable data to educators in real-time.

5.4 Limitations and Future Directions

While this review highlights the transformative potential of AI in educational settings, several limitations in the current body of literature must be acknowledged. Primarily, there is a significant English-language bias, with the majority of studies conducted in Western contexts or focusing on English-language instruction. Furthermore, the lack of standardized outcome measures across the literature complicates the ability to perform robust meta-analyses. Many existing evaluations rely on short-term interventions, leaving a critical gap in our understanding of the long-term pedagogical impacts of these technologies.

To address these gaps, future research must move beyond exploratory case studies toward rigorous Randomized Controlled Trials (RCTs) that utilize standardized assessment frameworks. Beyond purely cognitive or academic gains, there is a pressing need to investigate affective outcomes. As noted by [26], dimensions such as learner engagement, disciplinary identity, and self-efficacy are crucial for holistic development but remain under-researched in the context of AI-mediated learning.

Additionally, the practical barriers to institutional adoption remain a significant hurdle. Future studies should prioritize the development of cost-effectiveness models to provide school administrators with the data necessary for sustainable implementation. Finally, the rapid integration of AI in classrooms necessitates a deep dive into algorithmic ethics. Research must proactively address concerns regarding data privacy, the mitigation of algorithmic bias, and the potential for these tools to inadvertently widen the digital divide among marginalized student populations.

6 Conclusion: Toward an Inclusive Learning Future

This systematic review and meta-analysis represent the first comprehensive quantitative synthesis of AI-based tutoring systems for Deaf and Hard-of-Hearing (DHH) learners, encompassing 18 empirical studies and 1,245 participants. Our findings demonstrate that AI-powered educational technologies are not merely experimental curiosities but viable, evidence-based tools for advancing educational equity. The pooled effect size of $g = 0.48$ confirms a statistically significant, moderate positive impact on academic outcomes; a finding that should inform both educational policy and technological development.

Three critical insights emerge from our analysis. First, the strongest effects were observed in literacy and language acquisition domains ($g = 0.55$), suggesting that AI systems are particularly valuable where communication barriers are most pronounced. Second, systems designed with sign language as the primary communication mode ($g = 0.61$) significantly outperformed those relying on spoken language interfaces, validating the importance of culturally and linguistically responsive design. Third, system maturity proved consequential: commercial-grade platforms ($g = 0.60$) delivered substantially greater benefits than research prototypes ($g = 0.38$), emphasizing that pedagogical integration and user experience design are as important as the underlying AI algorithms.

These findings carry significant implications for multiple stakeholders. For **educators and administrators**, they provide empirical justification for investing in AI-based tools that support bilingual/bicultural approaches to DHH education. For **researchers and developers**, they highlight promising directions: integrating continuous sign language recognition with adaptive scaffolding, focusing on literacy development, and transitioning from proof-of-concept prototypes to robust, scalable platforms. For **policymakers**, they underscore the need for funding mechanisms that support both technological innovation and implementation science.

However, our work also reveals important limitations and future directions. The substantial heterogeneity ($I^2 = 67\%$) across studies indicates that not all AI implementations are equally effective, and more standardized evaluation frameworks are needed. Long-term studies are scarce, leaving questions about the sustainability of learning gains. Additionally, most research has focused on cognitive outcomes, with affective and social-emotional dimensions (crucial for holistic development) remaining underexplored.

Looking forward, we envision a learning landscape where AI serves not as a replacement for human interaction but as an amplifier of human potential. Future systems should embrace a *participatory design* ethos, actively involving DHH learners, educators, and community members throughout the development process. They should prioritize *multimodal intelligence*, seamlessly integrating computer vision for sign recognition, natural language processing for text generation, and affective computing for emotional support. Most importantly, they must embody an *ethics of care*, ensuring that technological advancement does not come at the cost of privacy, autonomy, or cultural integrity.

As AI capabilities continue to evolve at an unprecedented pace, our commitment to educational inclusion must evolve in tandem. The “silent divide” is not inevitable since it is a product of historical neglect and inadequate design. This meta-analysis demonstrates that AI-based tutoring systems can help bridge this divide, but their success depends on our collective will to center the needs, languages, and aspirations of DHH learners. By doing so, we can transform classrooms from spaces of accommodation to ecosystems of empowerment, where every learner (regardless of hearing status) can access, engage with, and excel in their educational journey.

In closing, we affirm that the most intelligent systems are not those with the most sophisticated algorithms, but those that most effectively honor human diversity. The future of AI in DHH education lies not in creating perfect machines, but in designing thoughtful partnerships between technology and pedagogy, between innovation and equity, and ultimately, between possibility and personhood.

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Chapter 2

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4	Fig1	<pre> graph TD A[Identification 500 records from 5 databases] --> B[Screening 450 after duplicates] B -.-> C[Excluded: 350 irrelevant] B --> D[Eligibility 100 full-text reviewed] D -.-> E[Excluded: 82 no AI/no outcomes] D --> F[Included 18 studies (1,245 participants)] </pre>	A flow chart showing the process of selecting studies for review. It starts with identifying 500 records from five databases. After removing duplicates, 450 records remain for screening, of which 350 are excluded as irrelevant. Then, 100 full-text articles are reviewed for eligibility, with 82 excluded for lacking artificial intelligence content or outcomes. Finally, 18 studies with a total of 1,245 participants are included. The chart highlights the narrowing down of studies through screening and eligibility checks to reach the final included studies.
8	Fig2		This chart compares the effect sizes of 15 individual studies on AI tutoring's impact on learning outcomes for deaf and hard of hearing students, shown as blue dots with horizontal lines representing confidence intervals. The overall combined effect is highlighted as an orange diamond near the top, indicating a moderate positive effect size around 0.5. The chart shows that most studies report positive effects, though the range of effect sizes and confidence intervals varies. This visual helps understand the general positive influence of AI tutoring across multiple studies, emphasizing the consistency and strength of the overall result.